RCC INSTITUTE OF INFORMATION TECHNOLOGY

DEPT OF ELECTRONICS AND COMMUNICATION ENGINEERING

CA-1 PRESENTATION

- NAME: Sk Sumit Ahmed
- YEAR: 3rd
- SEM: 6th
- CLASS ROLL NUMBER: ECE2023065
- UNIVERSITY ROLL NUMBER: 11700323087
- PAPER NAME: Computer Network
- PAPER CODE: EC602

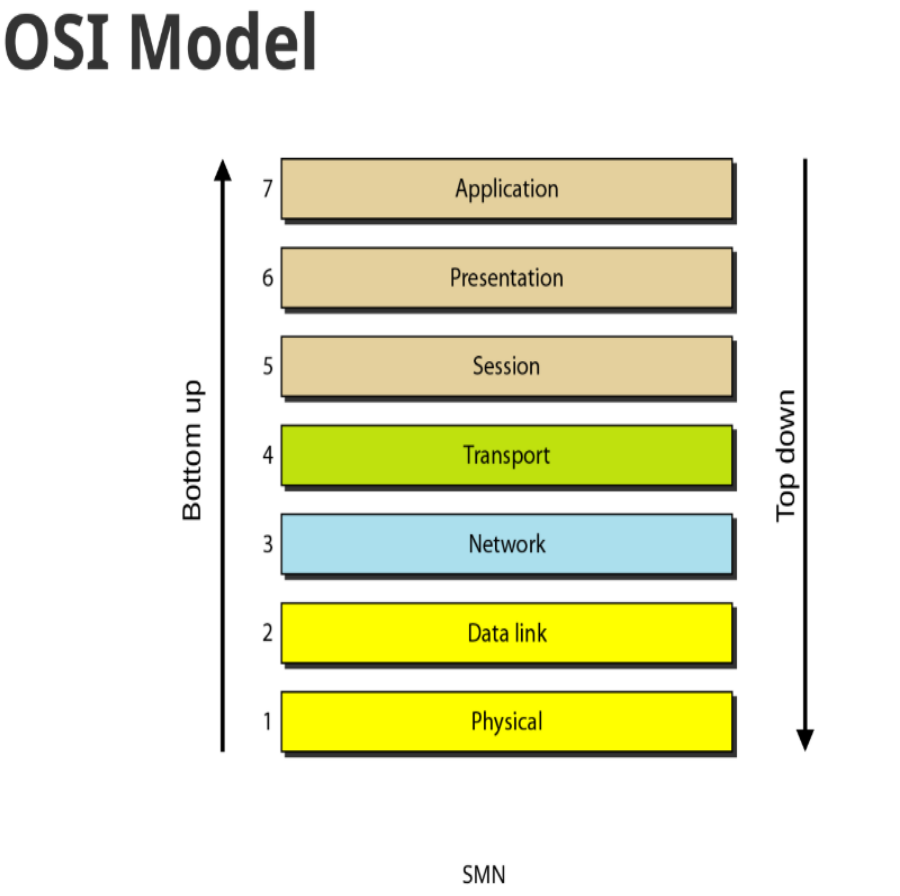
OSI layer structure

Content

- **Overview:** What is the OSI Model?
- **The Layered Approach:** Why do we split it up?
- **The 7 Layers:** A Top-Down Walkthrough.
- **Real-World Analogy:** The "Letter Writing" Concept.
- **Data Flow:** Encapsulation & Decapsulation.
- **Comparison:** OSI vs. TCP/IP.
- **Conclusion.**
- **References.**

What is the OSI Model?

•**Definition:** The Open Systems Interconnection (OSI) model is a theoretical framework that describes how data flows between computers.
•**Purpose:**
•It divides the complex problem of networking into **seven smaller, manageable problems**.
•This separation is called **"Layering"**.
•It serves as a standard so different computer systems can communicate.



The "Letter Writing" Analogy (Upper Layers)

Content: To understand the layers, let's use the analogy from our lecture notes: **Sending a Letter**

- **Layer 7: Application** -> **"You write the letter."**
  - This is the user interaction phase
- **Layer 6: Presentation** -> **"You write in English."**
  - This ensures the format (language/syntax) is understood by the receiver
- **Layer 5: Session** -> **"You arrange a meeting."**
  - This establishes the start and end of the communication

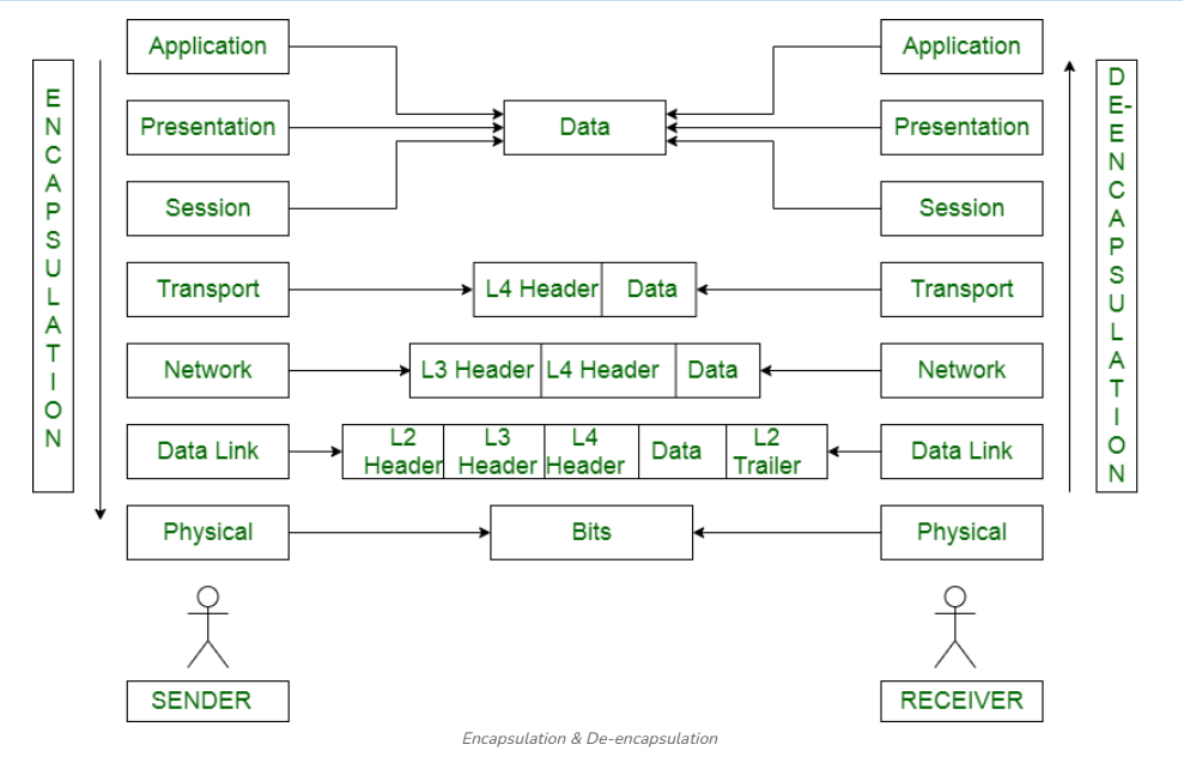
The "Letter Writing" Analogy (Lower Layers)

- **Layer 4: Transport** <- **"You choose the delivery speed."**
- Deciding between fast delivery (UDP) or safe/registered delivery (TCP)[9]
- **Layer 3: Network** <- **"The postal system finds the route."**
- Routing the letter to the correct address
- **Layer 2: Data Link** <- **"The postal worker ensures delivery."**
- Local delivery to the specific mailbox
- **Layer 1: Physical** <- **"The letter moves via road or air."**
- The actual physical movement of data (cables, Wi-Fi)

How Data Moves (Encapsulation)

- 🔹 **Going Down:** As data moves from Application to Physical, each layer adds a **"Header"** (meta-data). This is like putting the letter in an envelope, then stamping it.

- **Going Up:** When the receiver gets it, they strip away these headers layer by layer to get the original message.



Comparison - OSI vs. TCP/IP

While OSI is 7 layers, the practical model we use for the Internet is **TCP/IP** (4 Layers)

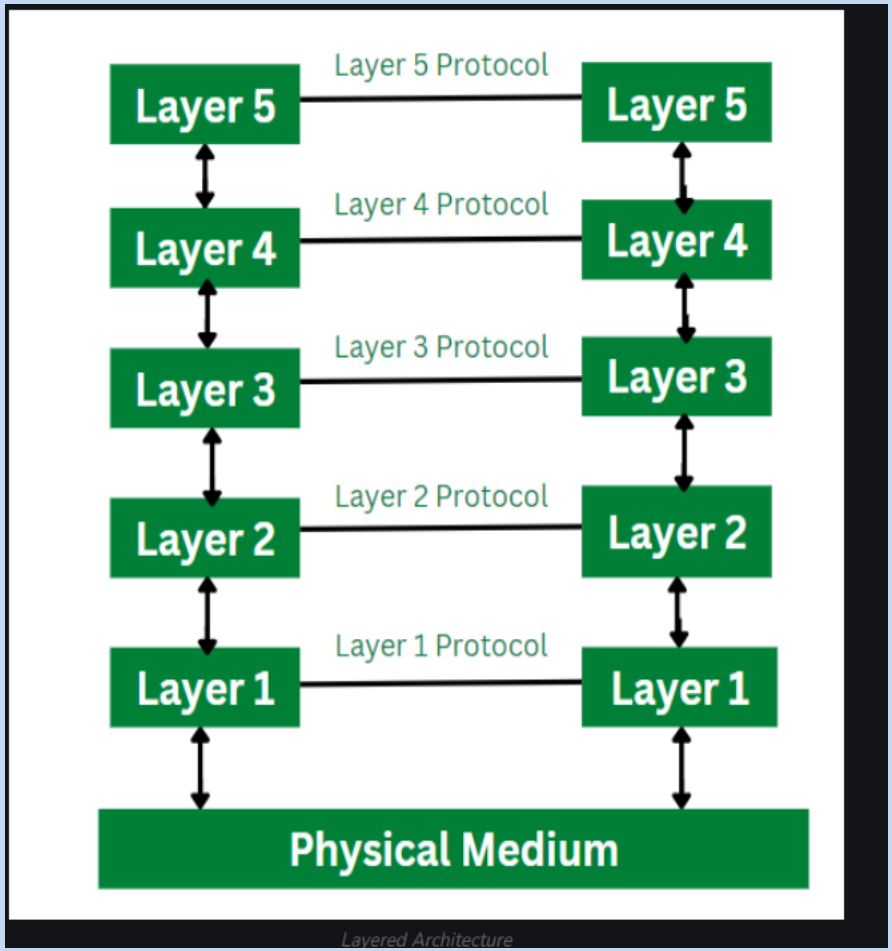
Feature	**OSI Model**	**TCP/IP Model**
Layers	7 Layers	4 Layers
Focus	Theoretical / Learning	Practical / Real-World
Usage	Understanding concepts	Actual Internet (HTTP, IP)

Why Use a Layered Architecture?

•**Simplicity:** It breaks networking into smaller pieces that are easier to design and troubleshoot.

•**Flexibility:** If we want to upgrade the hardware (Physical Layer), we don't have to rewrite the Application software (Layer 7).

•**Debugging:** It's easier to find errors. (e.g., Is the cable broken? Or is the server down?) .



Conclusion

•The **OSI Model** provides the universal standard for how different systems communicate.
•It moves data from the **Application** (Software) down to the **Physical** (Hardware) using specific protocols at every step.
•**Key Takeaway:** "OSI is for learning how the network works; TCP/IP is for actually building the network".

References

•**Lecture Notes:** Dr. Sujoy Mondal, Dept. of ECE, RCCIIT - *"Computer Network: Introduction"*.
•**Textbook:** B. A. Forouzan, *"Data Communications and Networking"* (3rd Ed.), TMH.
•**Textbook:** W. Stallings, *"Data and Computer Communications"* (5th Ed.), PHI.
•**Online Resource:** NPTEL Course - *Computer Networks and Internet Protocol*, IIT Kharagpur.